

From Eqn. (2), $c_2' = c_3' \left(\frac{a_1 - 3b_1}{17} \right)$

From Eqn. (1), $c_1' = c_3' \left(\frac{b_1 - 6a_1}{17} \right)$

Substituting the value of c_1' , c_2' and c_3' in the eqn. $c_1'u + c_2'v + c_3'w = 0$

$$\cancel{c_3' \left(\frac{b_1 - 6a_1}{17} \right) u} + \cancel{c_3' \left(\frac{a_1 - 3b_1}{17} \right) v} + c_3' \left(\frac{b_1 - 6a_1}{17} \right) \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + c_3' \left(\frac{a_1 - 3b_1}{17} \right) \begin{bmatrix} 1 \\ 6 \\ 0 \end{bmatrix} + c_3' \begin{bmatrix} a_1 \\ b_1 \\ c_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \left(\frac{b_1 - 6a_1}{17} \right) \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + \left(\frac{a_1 - 3b_1}{17} \right) \begin{bmatrix} 1 \\ 6 \\ 0 \end{bmatrix} + \begin{bmatrix} a_1 \\ b_1 \\ c_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

($\because c_3' \neq 0$)

$$\therefore \left(\frac{b_1 - 6a_1}{17} \right) \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + \left(\frac{a_1 - 3b_1}{17} \right) \begin{bmatrix} 1 \\ 6 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 + (-a_1) \\ 0 + (-b_1) \\ 0 + (-c_1) \end{bmatrix} \quad (\text{Vector addition})$$

$$\therefore \left(\frac{b_1 - 6a_1}{17} \right) \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + \left(\frac{a_1 - 3b_1}{17} \right) \begin{bmatrix} 1 \\ 6 \\ 0 \end{bmatrix} = \begin{bmatrix} -a_1 \\ -b_1 \\ -c_1 \end{bmatrix}$$

$$\Rightarrow \frac{(6a_1 - b_1)}{17} \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + \frac{(3b_1 - a_1)}{17} \begin{bmatrix} 1 \\ 6 \\ 0 \end{bmatrix} = \begin{bmatrix} (-1) \cdot (-a_1) \\ (-1) \cdot (-b_1) \\ (-1) \cdot (-c_1) \end{bmatrix} \quad (\text{Scalar multiplication})$$

$$\Rightarrow \frac{(6a_1 - b_1)}{17} \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + \frac{(3b_1 - a_1)}{17} \begin{bmatrix} 1 \\ 6 \\ 0 \end{bmatrix} = \begin{bmatrix} a_1 \\ b_1 \\ c_1 \end{bmatrix}$$

$\begin{matrix} u & v & w \end{matrix}$

$\therefore$ By defn, $w \in \text{Span}\{u, v\}$ $\square$

Theorem 8: If a set contains more vectors than there are entries in each vector, then the set is linearly dependent. That is, any set $\{v_1, \dots, v_p\}$ in $\mathbb{R}^n$ is linearly dependent if $p > n$.

Proof: Let, $a_1, a_2, \dots, a_p$ be arbitrary real number weights. The set $\{v_1, v_2, \dots, v_p\}$ is linearly dependent if $a_1v_1 + a_2v_2 + \dots + a_pv_p = 0$, and not all the weights are 0 together. We need to find out whether such a solution in weights of the equation exists.

$$\begin{bmatrix} v_1 & v_2 & \dots & v_p \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_p \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}. \quad (\text{By defn. of matrix eqn. } Ax=b)$$